



Moolya Testing

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# Accelerating DCS System Quality through Scalable Test Automation

Proven Success in the DCS Industry Using Low-Code and Python-Based Frameworks

■ 14+

Years in Business

■ 200+

Customers

■ 20+

Countries

■ 80%

Industry leaders choose us

# Introduction & Objective

As industries move toward increased digitization, the need for robust and scalable test automation has become critical especially in complex domains like Industrial Automation and Distributed Control Systems (DCS). This presentation shares our experience implementing a successful test automation solution for a leading customer in the same domain. We will walk through the challenges they faced, our solution approach, and the tangible business value delivered. Our goal is to show how similar success can be achieved for your organization by leveraging proven practices and scalable frameworks.



# Customer Context & Ask

- Customer in Industrial Automation / DCS domain.
- Needed structured Smoke and Regression automation.
- Existing framework based on TestComplete with VBScript.
- Struggled with scalability and traceability.

# Challenges Faced

- VBScript-based automation was hard to scale.
- No mapping between test cases and scripts.
- Manual testing efforts still significant.
- No consistent execution or reporting process.

# Our Solution Overview

- Assessed existing framework and gaps.
- Proposed new framework using TestComplete + Python.
- Introduced low-code/no-code interface for faster test creation.
- Focused on automating critical Smoke test cases first.
- Categorized test cases as Simple, Medium, Complex.
- Successfully automated complex user flows.

# Framework Architecture

- Modular architecture with TestComplete + Python.
- Low-code interface for ease of use.
- Mapping of test cases to scripts via metadata.
- Setup for multi-system execution and scalability.
- CI/CD readiness for future integration.

# Execution & Reporting Strategy

- Daily execution across multiple systems.
- Execution matrix for Smoke, Regression, and Feature-specific suites.
- Regular demo cadence with stakeholders.
- Automated reporting with logs, screenshots, and summary in HTML/Excel/PDF.

# Business Impact **Delivered**

- 50% faster script creation using low-code tooling.
- Improved traceability and maintainability.
- Achieved 40% automation coverage in Phase 1.
- Automated complex and business-critical flows.
- Enabled continuous feedback with daily runs.
- Increased confidence in release quality.

# Quantified ROI & Effort Savings

- Achieved 60% effort savings in regression validation cycles.
- Early test automation reduced defect leakage by 30%.
- Significant reduction in manual QA effort across releases.
- Improved test productivity with leaner QA bandwidth.

# Testimonials & Case Reference

- Feedback from the existing industrial automation client:
  - “The automation coverage achieved in a short span helped us reduce our QA cycle time significantly.”
  - “The team understood our complex workflows and delivered results with high reliability.”
- Optional NDA-safe case study summary

# Why This Matters to You

- Deep understanding of Industrial Automation/DCS complexities.
- Proven success in automating domain-specific workflows.
- Scalable, maintainable framework ready to adapt.
- Faster automation ROI using low-code tooling.
- Ready to replicate and customize success for your needs.